INTERNSHIP PROJECT REPORT

Network Scanner Using Python

Submitted by: Samyak Bhaisare
Year: 2026

1. Introduction

Network scanning is an important technique used in cybersecurity and networking to identify active devices connected within a network. This project implements a multithreaded network scanner using Python and Scapy to discover devices, detect MAC addresses, and resolve hostnames.

2. Objectives

- Detect active devices in a network
- Identify IP and MAC addresses
- Resolve hostnames
- Implement multithreading for faster scanning
- Save scan results into a text file

3. Tools and Technologies Used

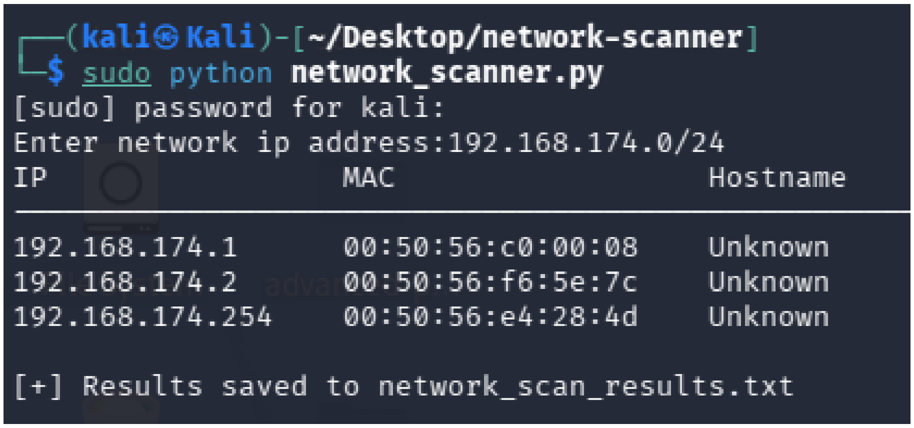
- Python
- Scapy
- Socket Programming
- Threading
- Queue
- VS Code / Kali Linux
- Npcap

4. Methodology

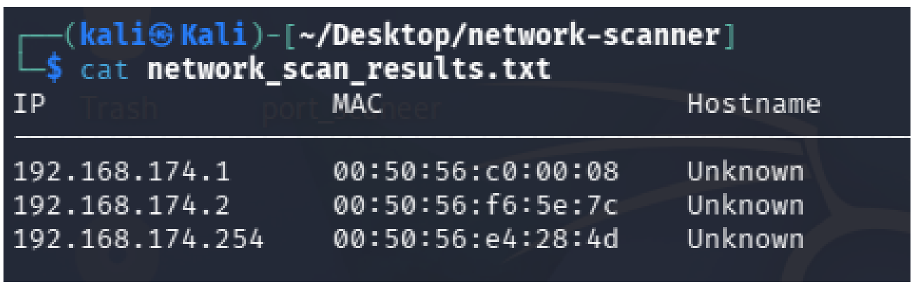
1. User enters a CIDR-based network range
2. ARP request packets are generated
3. Broadcast packets are sent in the network
4. Active devices respond with MAC addresses
5. Hostnames are resolved using socket programming
6. Results are displayed and saved into a text file

5. Results

The network scanner successfully detected active devices connected to the network along with their MAC addresses and hostnames. The scan results were also saved into a text file for future reference.



Caption: Figure 1 – Network Scanner Output



Caption: Figure 2 – Saved Results File

6. Applications

- Network monitoring
- Device discovery
- Cybersecurity learning
- Basic network analysis

7. Ethical Considerations

This tool should only be used on authorized networks and systems for educational and ethical purposes.

8. Conclusion

The project successfully demonstrates ARP-based network scanning using Python and Scapy. It provides practical understanding of networking concepts, packet communication, and multithreading. The tool can be further enhanced with advanced scanning and monitoring features in the future.